Week of 3/3 and 3/5:

Tuesday 3/3:

We started our new lab on Quantum Key Distribution. We read about the hardware we are using and the procedure we need to complete for our lab. We read through the intro materials on the canvas and looked at the matlab code that we will be using for this experiment. We made sure we could use the equipment and that the code worked properly.

Thursday 3/5:

We continued to learn about Quantum Key Distribution and did more research on what our lab requires us to do. We also coded up a way to convert images into a bit string.

Our figure demonstrates the process of converting an image to a bit-string, and reverting a bit-string to an image. (a) represents our original image (with a border added). (b) represents the grayscale conversion of our image, assigning a value in [0,1). (c) represents the conversion of this matrix into a matrix of 8-bit numbers, by multiplying each grayscale value by 256 and rounding. We then flatten the array, transform every number into binary, and append them all into a bitstring (d). We undo this process by splitting our bit-string into 8-bit chunks, converting each chunk to decimal, then reshaping our array back into a matrix. We recover our original grayscale image in (e).

The next steps with this process are to consider ancillary data (to allow us to reconstruct our image from noisy data), write algorithms to detect bit-flip errors, and put everything together for our initial QKD communication.